Databases and corpora 2

Quantitative Methods in Historical Linguistics

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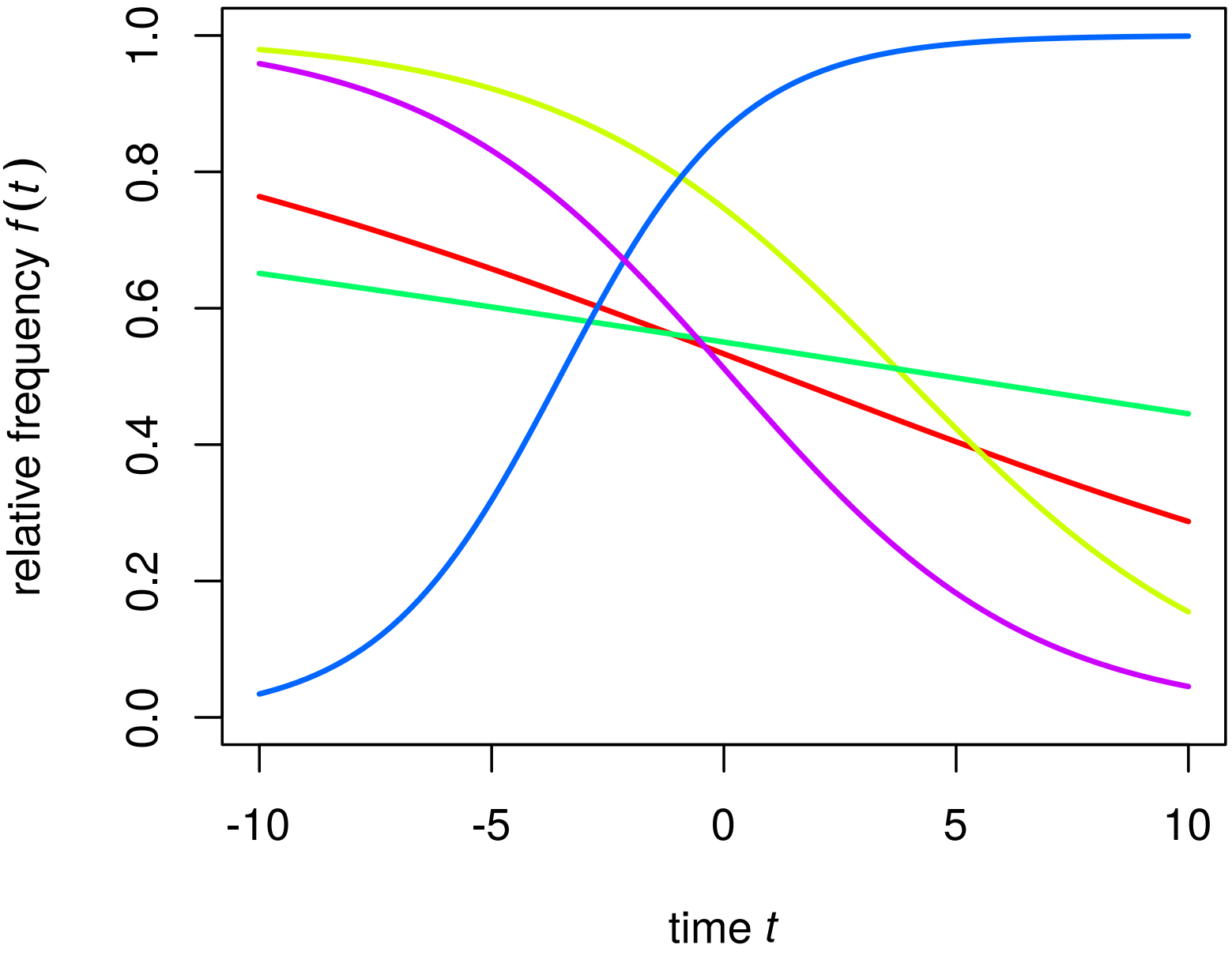
First things first

- Questions?
- R plotting tutorial delayed (sorry!)
- Please download the following datasets from ILIAS:
  - `ellegard.csv` (same as last time)
  - `ellegard-full.csv`
- Discuss course evaluation


relative frequency $f(t)$
0.0
0.2
0.4
0.6
0.8
1.0
-10
-5
0
5
10
time t

$$f(t) = \frac{1}{1 + \exp(s \times (k - t))}$$

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- t is the **VARIABLE** (time)
- s and k are **PARAMETERS**/**CONSTANTS** (slope and intercept)

Logistic functions

- With s and k, we have defined a **FAMILY** of S-curves
- The parameter k controls the…

Logistic functions

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  - For positive k, the change is shifted…

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- The parameter k controls the…**time position of the change**
  - For positive k, the change is shifted…**to the right**
  - For negative k, the change is shifted…

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- The parameter k controls the…**time position of the change**
  - For positive k, the change is shifted…**to the right**
  - For negative k, the change is shifted…**to the left**
- The parameter s controls the…

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  - For negative k, the change is shifted…**to the left**
- The parameter s controls the…**steepness of the change**
  - For positive s, the relative frequency…

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  - For positive s, the relative frequency…**increases over time**
  - For negative s, the relative frequency…

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  - The larger the absolute value $|s|$, the…

Logistic functions

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  - For positive s, the relative frequency…**increases over time**
  - For negative s, the relative frequency…**decreases over time**
  - The larger the absolute value $|s|$, the…**steeper the change**
- s is known as the **SLOPE**, k the **INTERCEPT**
- We can now try to precisely describe patterns in historical data, by finding out which combination of s and k gives the best fit!

Curve-fitting

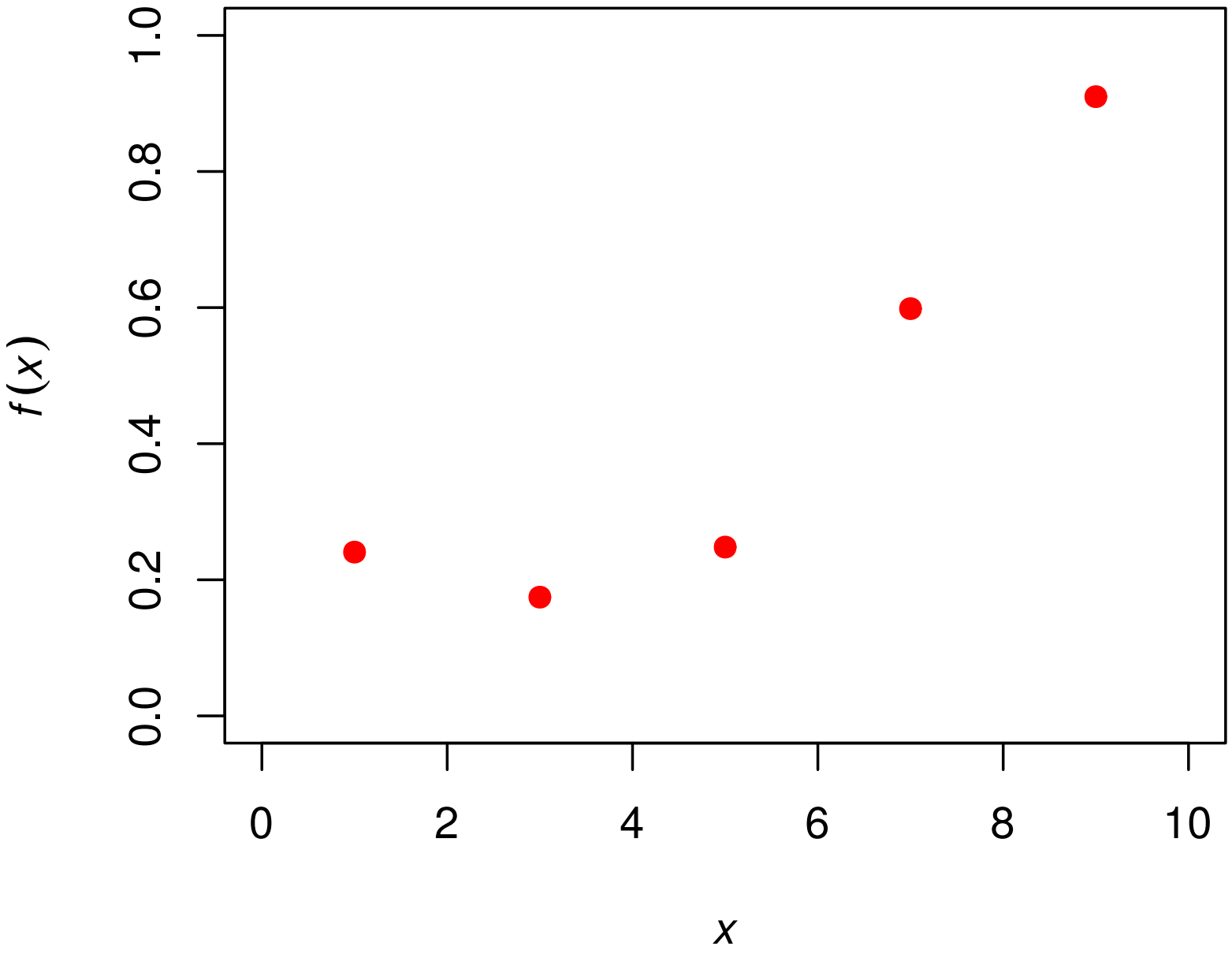
- Our goal today is to describe some actual historical changes using these S-curves
- I.e. we will be concerned with **CURVE-FITTING**

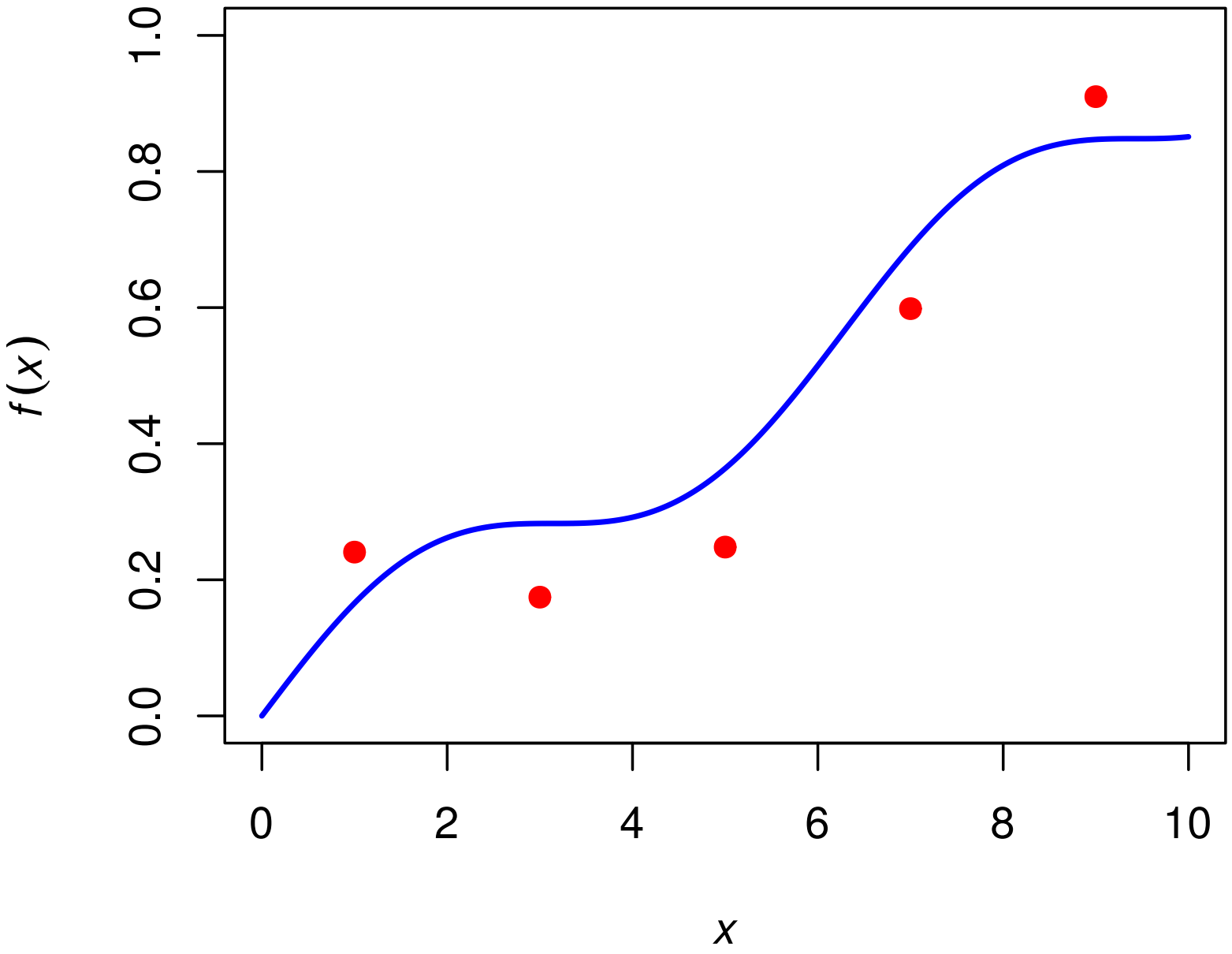
The logistic curve-fitting problem

Given a data set D, how do we find the combination of s and k parameters that yields the best fit to D?

Sub-problem

How can we measure the goodness of a fit?


1.0
0.8
0.6
0.4
0.2
0.0
f(x)
0
2
4
6
8
10
x


1.0
0.8
0.6
0.4
0.2
0.0
f(x)
0
2
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6
8
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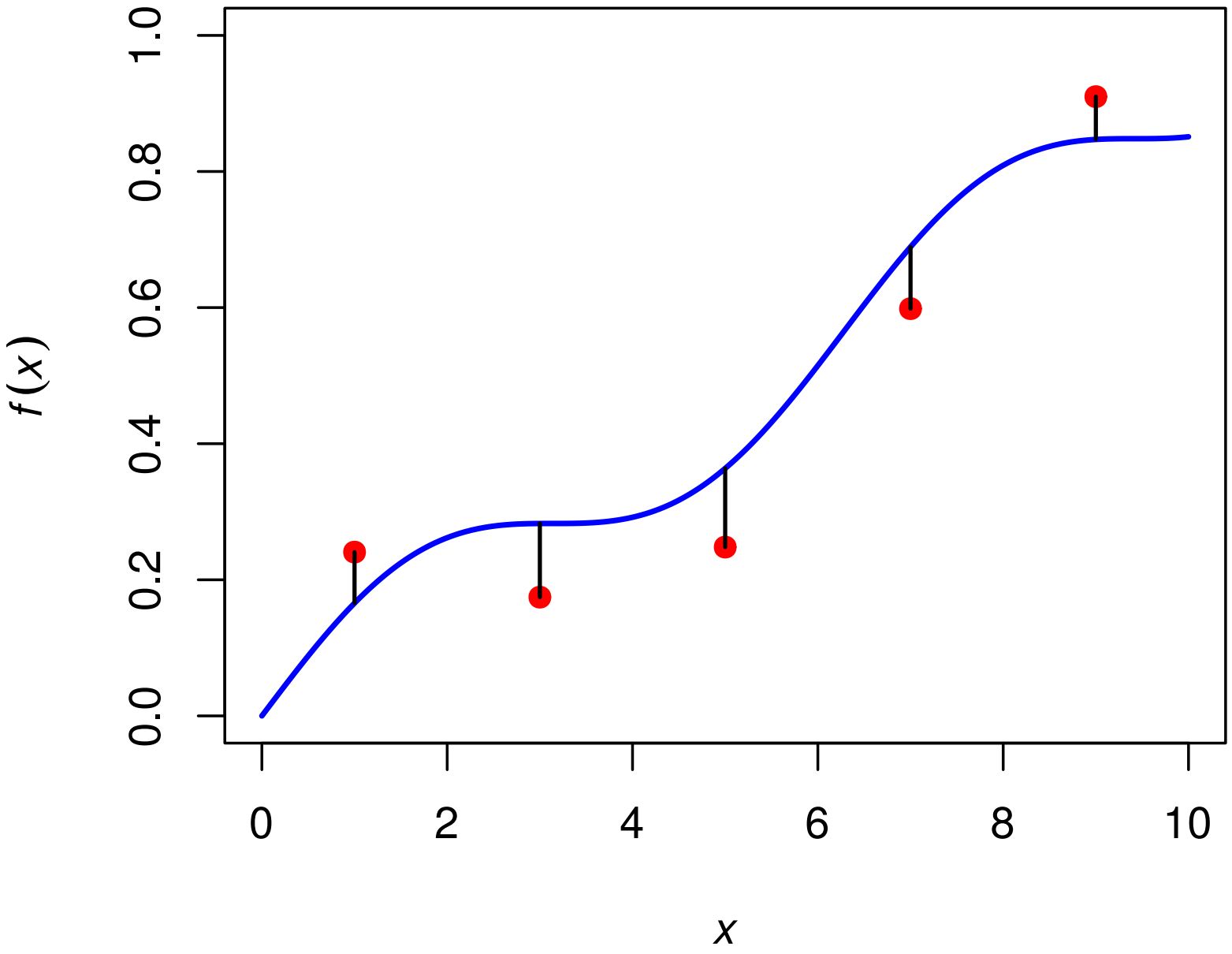
x


1.0
0.8
0.6
0.4
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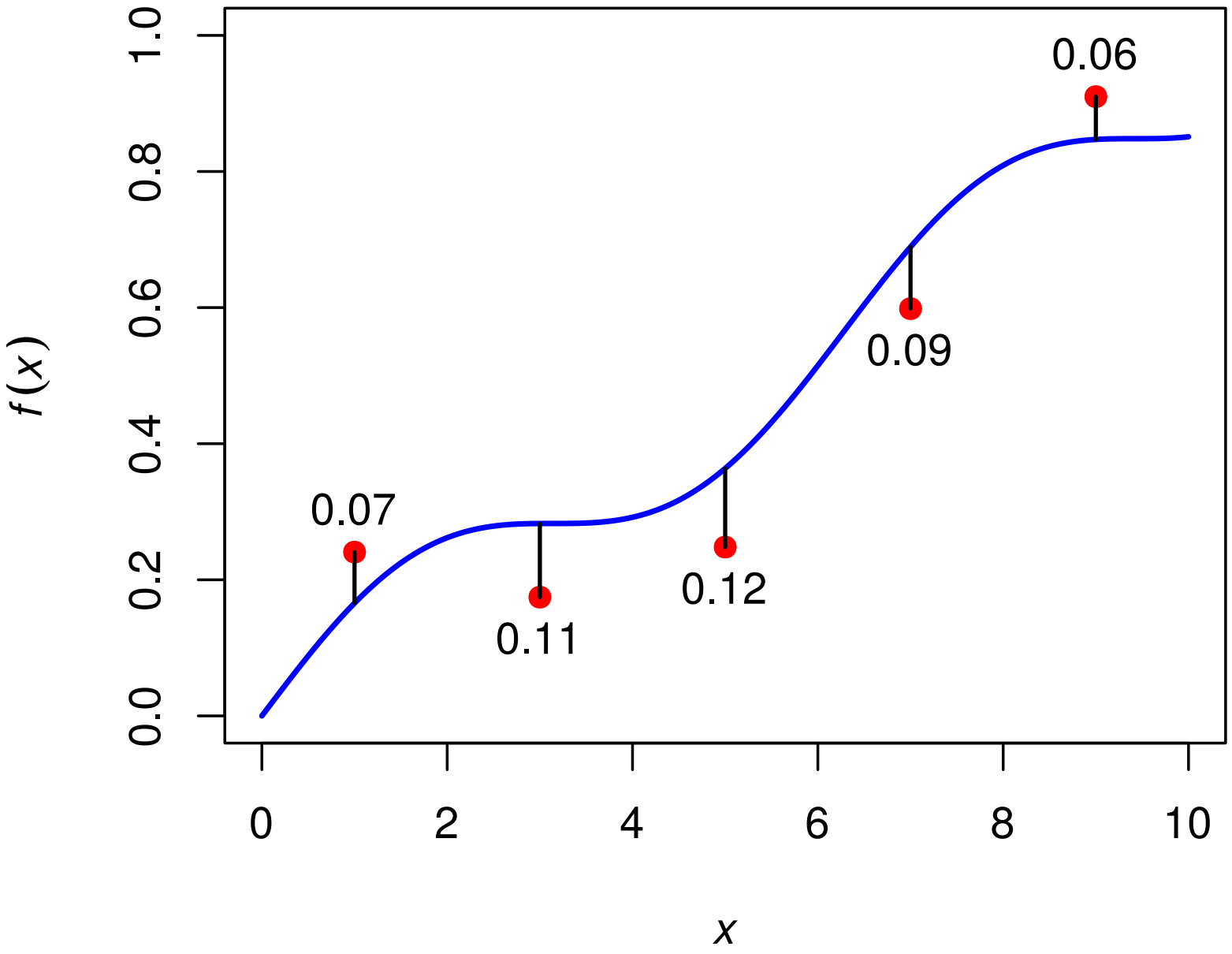
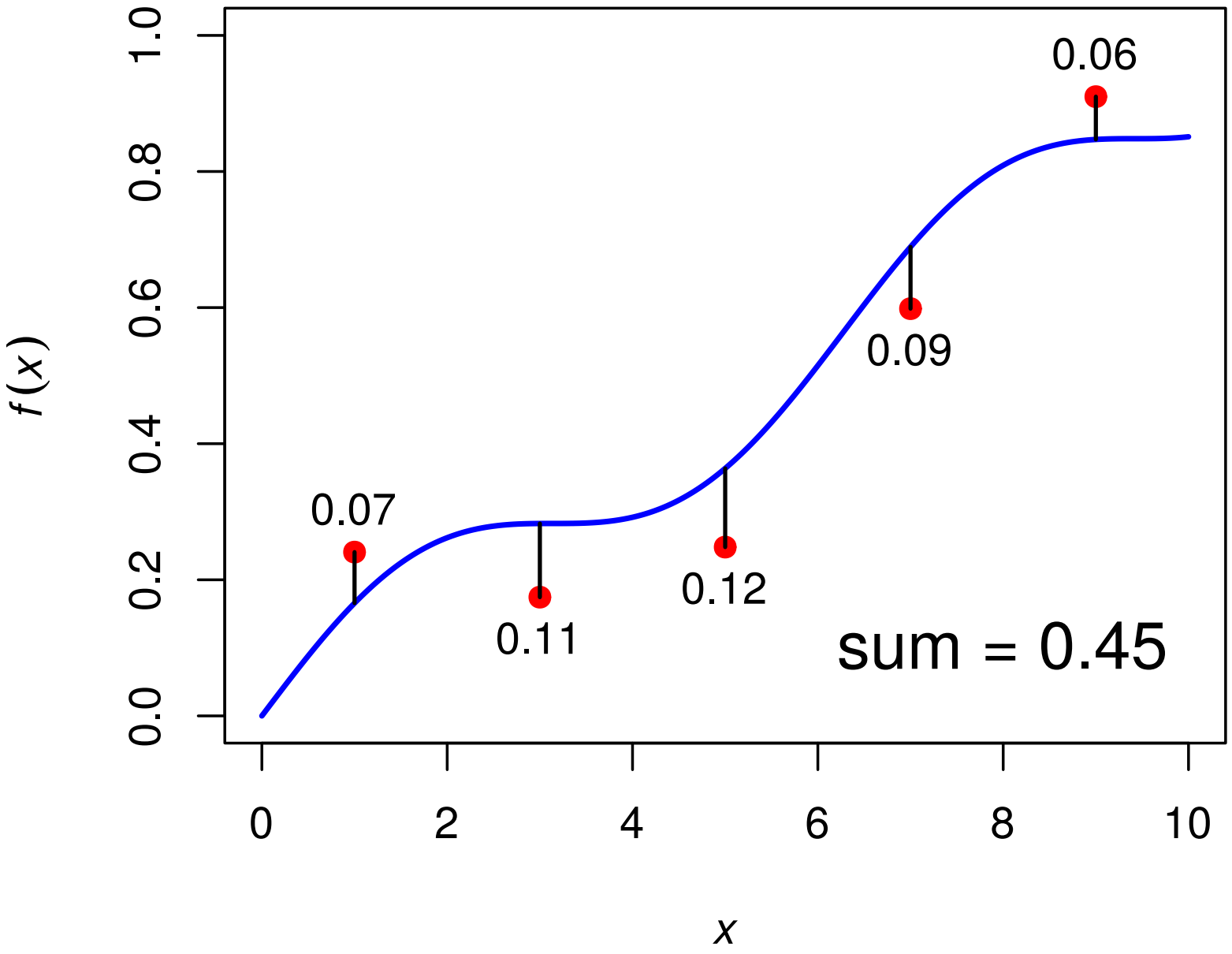
0.0
f(x)
0
2
4
6
8
10
x


0.06
0.09
0.07
0.11
0.12
1.0
0.8
0.6
0.4
0.2
0.0
f(x)
0
2
4
6
8
10
x


0.06
0.09
0.07
0.11
0.12
sum = 0.45
1.0
0.8
0.6
0.4
0.2
0.0
f(x)
0
2
4
6
8
10
x

Error

- Each data point is a pair (x_i, y_i)
- And for each x_i, the curve gives a prediction $f(x_i)$
- Distance between the two is $|y_i - f(x_i)|$
- Error of fit can be defined as:

$$\sum_i |y_i - f(x_i)|$$

  (This gave the value **0.45** on the previous slide)
- Often people use:

$$\sum_i (y_i - f(x_i))^2$$

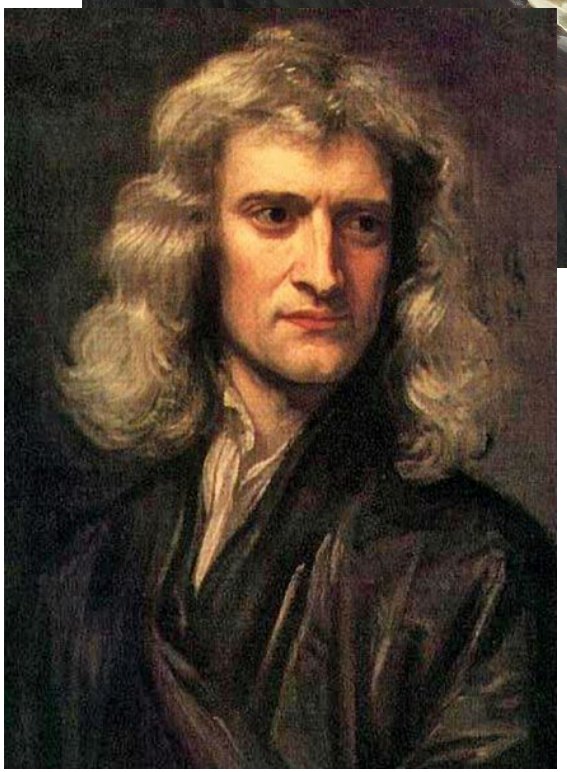
  known as the **SUM OF SQUARED DIFFERENCES**
- The closer to zero the better!

Least squares optimization

- A **LEAST SQUARES ALGORITHM** can be used to find the best fit
- (By its name, such an algorithm tries to minimize the sum of squared differences)
- This is easy if the function is linear i.e. a straight line
- But very difficult if it is anything else, such as an S-curve
  - $\Rightarrow$ Let the computer do it
  - $\Rightarrow$ We use what are known as **NONLINEAR LEAST SQUARES** algorithms

Gauss–Newton algorithm

- The classical nonlinear least squares algorithm
- Fast, but **NOT** guaranteed to always find the best fit!
  - May find a suboptimal fit
  - May find no fit at all
  - Sensitive dependence on initial conditions (next slide…)



Gauss–Newton algorithm

- Applied to our logistic case, this algorithm
  1. Takes an initial guess of the parameters s_0 and k_0
  2. Calculates the sum of squares
  3. Takes a better guess s_1 and k_1
  4. Again calculates the sum of squares
  5. Takes an even better guess s_2 and k_2

  … And so on and so forth until the error cannot be made any smaller
- Next we look at some fancy animations…(ILIAS folder 'Other stuff')

Gauss–Newton in R

- To do Gauss–Newton in R, we need a data frame and a **FORMULA**, an R expression that gives the curve we want to fit
- Examples of formulae:

formula	curve
`y~x`	$y = x$
`y~x^2`	$y = x^2$
`y~a*x+b`	$y = ax + b$

Reflection Exercise

What is the R formula corresponding to the logistic curve?

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Reflection Exercise

What is the R formula corresponding to the logistic curve?

```
y~1/(1 + exp(s*(k-x)))
```

- The fitting itself is done by a command called `nls` (short for nonlinear least squares)

Gauss–Newton in R

- Recall Ellegård's data on English *do*-support:

```
head(elleg)

##      period start  end AQ_do AQ_nodo
## 1 1400-1425  1400 1425     0      10
## 2 1425-1475  1425 1475     6     136
## 3 1475-1500  1475 1500    10     132
## 4 1500-1525  1500 1525    41     140
## 5 1525-1535  1525 1535    33      69
## 6 1535-1550  1535 1550    93     114
```

- Last time we added a column containing the relative frequency of *do*-support (in questions) and another column containing time period midpoints
- Please do that again now (or load your data frame from last time)

Gauss–Newton in R

- Here's one way of doing it:

```
elleg$rel_do <- elleg$AQ_do/(elleg$AQ_do + elleg$AQ_nodo)
elleg$mids <- (elleg$start + elleg$end)/2
head(elleg)

##      period start  end AQ_do AQ_nodo     rel_do   mids
## 1 1400-1425  1400 1425     0      10 0.00000000 1412.5
## 2 1425-1475  1425 1475     6     136 0.04225352 1450.0
## 3 1475-1500  1475 1500    10     132 0.07042254 1487.5
## 4 1500-1525  1500 1525    41     140 0.22651934 1512.5
## 5 1525-1535  1525 1535    33      69 0.32352941 1530.0
## 6 1535-1550  1535 1550    93     114 0.44927536 1542.5
```

Gauss–Newton in R

- We now take the formula for the logistic S-curve:

```
y~1/(1 + exp(s*(k-x)))
```

- And replace `y` with `rel_do` and `x` with `mids`:

```
rel_do~1/(1 + exp(s*(k-mids)))
```

  (If you have differently named columns, use your column names)

- Why do we do this? Because the time period midpoints are on our *x* axis, while the relative frequencies are on our *y* axis

Gauss–Newton in R

- Then actually fit the curve:

```
fit <- nls(rel_do~1/(1 + exp(s*(k-mids))), elleg,
           start=list(s=0.1, k=1500))
```

- The arguments are:
  1. The curve formula
  2. The data frame
  3. A list of initial guesses for the parameter values
- Each of these arguments needs to be specified!

Gauss–Newton in R

- If all went well, `fit` now contains a nonlinear fit. To see its details, simply type `fit`:

```
fit

## Nonlinear regression model
##   model: rel_do ~ 1/(1 + exp(s * (k - mids)))
##    data: elleg
##         s         k
## 2.088e-02 1.566e+03
##  residual sum-of-squares: 0.03778
##
## Number of iterations to convergence: 8
## Achieved convergence tolerance: 3.129e-06
```

- You'll see the estimated parameters and remaining error at the end of the algorithm, among other details

Gauss–Newton in R

- To get just the parameters, you can use `coef`:

```
coef(fit)

##             s             k
## 2.087716e-02 1.565967e+03

coef(fit)[1]

##           s
## 0.02087716

coef(fit)[2]

##        k
## 1565.967
```

- And `deviance` for the error:

```
deviance(fit)

## [1] 0.03778483
```

Gauss–Newton in R

- A word of warning: as I said above, Gauss–Newton is not guaranteed to work!
- May find a fit that is not the best possible fit
- May fail to find a fit at all
- The latter usually occurs if the initial parameter guess is not good enough:

```
nls(rel_do~1/(1 + exp(s*(k-mids))), elleg,
    start=list(s=-0.1, k=1300))

## Error in nls(rel_do ~ 1/(1 + exp(s * (k -
mids))), elleg, start = list(s = -0.1, : singular
gradient
```

- In such a case, try another guess (and then another, and…)!

Do-support again

- *Do*-support in fact affects a number of linguistic contexts:

Middle English	**Modern English**
Drinks John tea?	Does John drink tea?
Drinks John not tea?	Doesn't John drink tea?
John drinks not tea	John doesn't drink tea

⇒ Both "affirmative" and "negative" questions, plus negative declaratives (plus some others, but we won't be looking at them)

- Ellegård has data on these

Do-support again

- Now please load the dataset `ellegard_full.csv`:

```
elleg_full <- read.csv("ellegard_full.csv")
```

- Dataset has columns of the form `X_do` and `X_freq`:

X	**context**
AQ	affirmative questions
NQ	negative questions
ND	negative declaratives

Exercise

Use the Gauss–Newton algorithm (`nls`) to find the best-fitting logistic curves (s and k parameters) for each context.

Finally...

- No session next week!
- No new Portfolio Exercise (so you'll have more time to work on past PEs)
- Please answer ILIAS poll about Workshop timing (in the sidebar to the right, below the Calendar)
- Any questions?